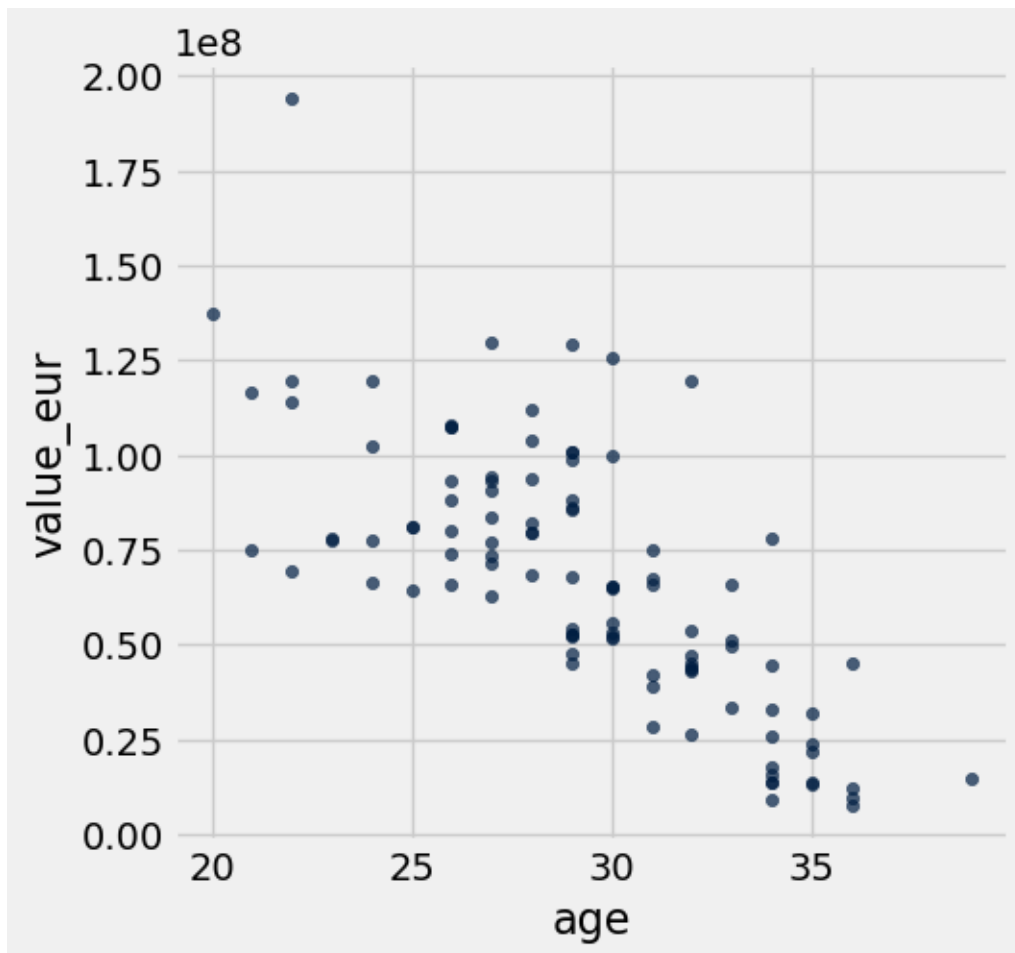


Question 2.1. Before jumping into any statistical techniques, it's important to see what the data looks like, because data visualizations allow us to uncover patterns in our data that would have otherwise been much more difficult to see. **(3 points)**

Create a scatter plot with age on the x-axis ("age"), and the player's value in Euros ("value_eur") on the y-axis.

```
In [62]: fifa.scatter("age", "value_eur")
```



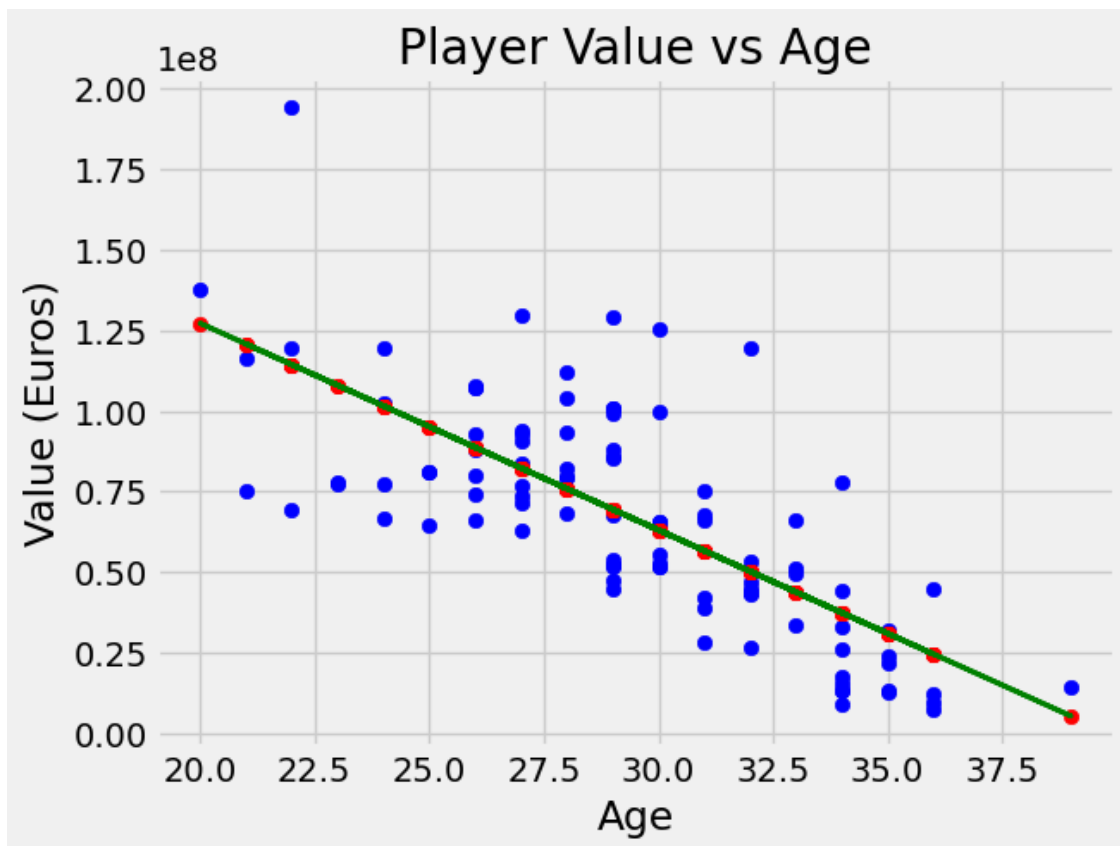
Question 2.3. Create a scatter plot with player age (“age”) along the x-axis and both real player value (“value_eur”) and predicted player value along the y-axis. The predictions should be created using a fitted **regression line**. The color of the dots for the real player values should be different from the color for the predicted player values. **(8 points)**

Hint 1: Feel free to use functions you have defined previously.

Hint 2: [15.2](#) and [7.3](#) has examples of creating such scatter plots.

```
In [65]: regression = np.polyfit(fifa.column("age"), fifa.column("value_eur"), 1)
         predictions = np.polyval(regression, fifa.column("age"))
         plt.scatter(fifa.column("age"), fifa.column("value_eur"), color = 'blue', label = 'Real Player
         plt.scatter(fifa.column("age"), predictions, color = 'red', label = 'Predicted Player Values')
         plt.plot(fifa.column("age"), predictions, color = 'green', linewidth = 2)
         plt.xlabel("Age")
         plt.ylabel("Value (Euros)")
         plt.title("Player Value vs Age")
```

```
Out[65]: Text(0.5, 1.0, 'Player Value vs Age')
```



Question 2.4. Looking at the scatter plot you produced above, is linear regression a good model to use? If so, what features or characteristics make this model reasonable? If not, what features or characteristics make it unreasonable? **(5 points)**

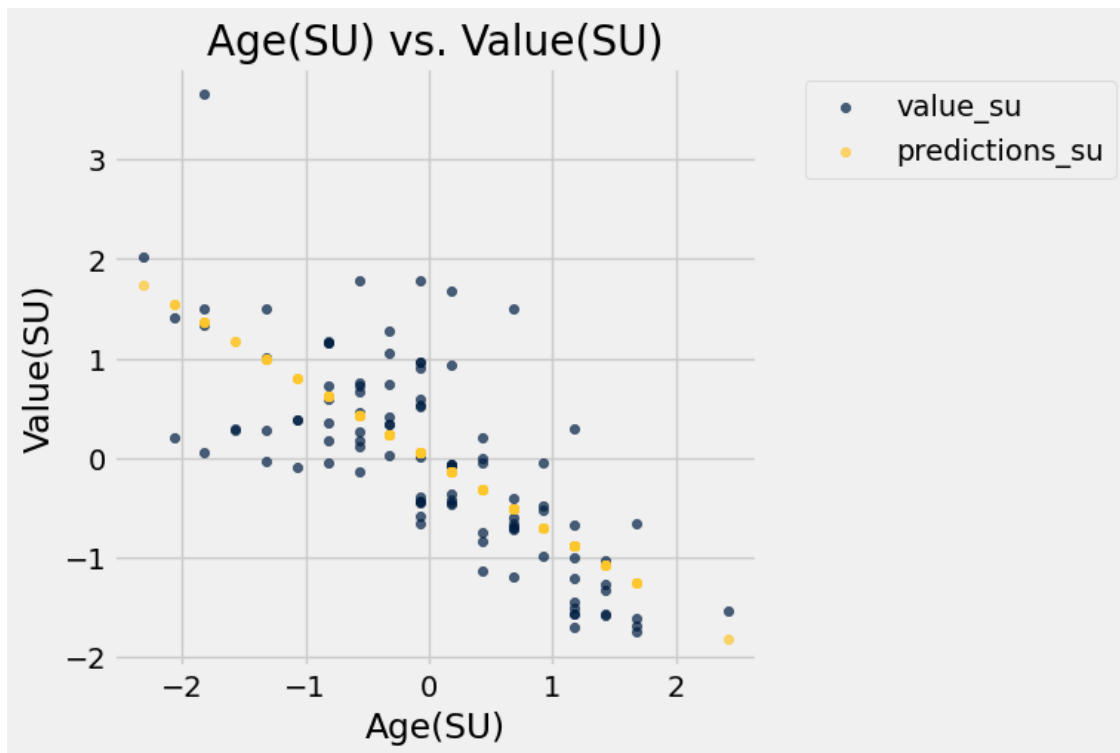
Looking at the scatter plot, there is a general negative trend between player age and player value. But, there is also a lot of variability in the data, with some points being further from the line of best fit than others, so linear regression may not be the best model to use.

Question 2.5. In 2.3, we created a scatter plot **in original units**. Now, create a scatter plot with player age **in standard units** along the x-axis and both real and predicted player value **in standard units** along the y-axis. The color of the dots of the real and predicted values should be different. **(8 points)**

Hint: Feel free to use functions you have defined previously.

```
In [66]: predictions_su = r_guess*standard_units(fifa.column("age"))
         fifa_su = Table().with_columns("age_su", standard_units(fifa.column("age")),
                                       "value_su", standard_units(fifa.column("value_eur")),
                                       "predictions_su", predictions_su)

         fifa_su.scatter("age_su")
         plt.xlabel("Age(SU)")
         plt.ylabel("Value(SU)")
         plt.title("Age(SU) vs. Value(SU)")
         plt.show()
```



Question 2.6. Compare your plots in 2.3 and 2.5. What similarities do they share? What differences do they have? **(5 points)**

In both plots, there is a negative relationship between value and age, and the points in both graphs are shaped similarly. However, the standard units graph contains negative values on the axes, which represent the values below the mean.

Question 2.8. Use the `rmse` function you defined along with `minimize` to find the least-squares regression parameters predicting player value from player age. Here's an [example](#) of using the `minimize` function from the textbook. (10 points)

Then set `lsq_slope` and `lsq_intercept` to be the least-squares regression line slope and intercept, respectively.

Finally, create a scatter plot like you did in 2.3 with player age ("age") along the x-axis and both real player value ("value_eur") and predicted player value along the y-axis. **Be sure to use your least-squares regression line to compute the predicted values.** The color of the dots for the real player values should be different from the color for the predicted player values.

Note: Your solution should not make any calls to the slope or intercept functions defined earlier.

Hint: Your call to `minimize` will return an array of argument values that minimize the return value of the function passed to `minimize`.

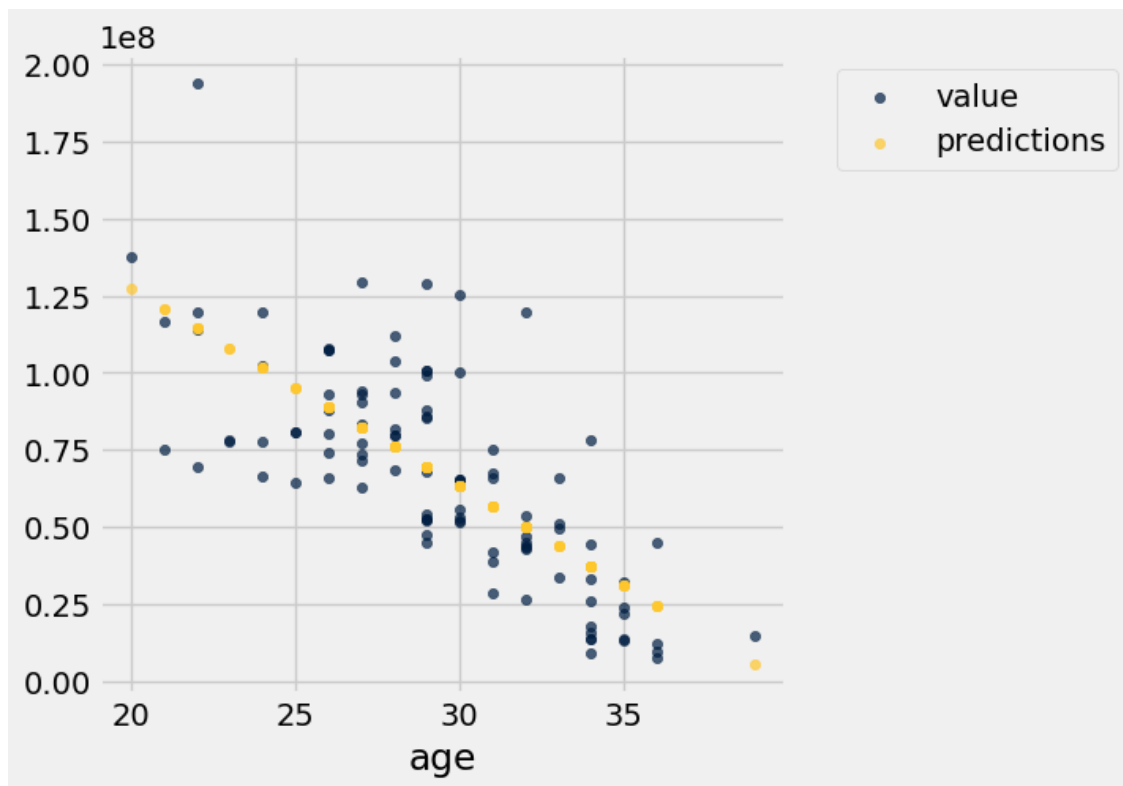
```
In [69]: minimized_parameters = minimize(rmse)
         lsq_slope = -6.41461552e+06
         lsq_intercept = 2.55525256e+08

         # This just prints your slope and intercept
         print("Slope: {:.g} | Intercept: {:.g}".format(lsq_slope, lsq_intercept))

         fifa_with_lsq_predictions = Table().with_columns("age", fifa.column("age"),
                                                         "value", fifa.column("value_eur"),
                                                         "predictions", predictions)

         fifa_with_lsq_predictions.scatter("age")
```

Slope: -6.41462e+06 | Intercept: 2.55525e+08



Question 2.9. The resulting line you found in 2.8 should appear very similar to the line you found in 2.3. Why were we able to minimize RMSE to find nearly the same slope and intercept from the previous formulas? (5 points)

Hint: Re-reading [15.3](#) might be helpful here.

Minimizing the RMSE limits the error between the observed values and the values predicted by the best fit line and allows us to plot a model that resembles the actual data much closer. A lower error means that the best fit line is more representative of the data, which is why we can see that the slope and intercept of the resulting line from 2.8 are similar to those of the line from 2.3.

